System Capacity & Care Load Analytics for Unaccompanied Children

Data Analytics Project Report

Capacity, care load, length of stay, early warning and 60-day forecast, with a Streamlit web application

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Data period analysed	23 Mar 2021 to 8 Oct 2026
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Project at a glance

Item	Result
Data	1,366 raw rows, daily, 23 Mar 2021 to 8 Oct 2026; HHS data for the full period, CBP data from 31 Jan 2024
Children in the system today	1,796 on 8 Oct 2026: 33 in CBP custody and 1,763 in HHS care
Peak HHS care	22,557 children on 29 Apr 2021; today's level is 7.8% of that peak
Capacity (assumption)	Threshold 12,336 children = 90th percentile of HHS care; utilization today 14.3%, peak 183%
Days above the threshold	203 days (10.0%), all in 2021
Estimated length of stay	About 30 days in 2021-2024; 123 days now (approximation, see 7.5)
Best forecast model	ARIMA(7,1,0): rolling-origin MAE 31.1 children vs 52.6 for the naive baseline
Forecast, 7 Dec 2026	1,704 children in HHS care (80% band 1,395 to 1,844), about 14% of the threshold
Early-warning rule	70.6% of alert days were followed by growth in HHS care within 14 days, against a 45.6% base rate

1. Background and Context

Unaccompanied children are minors who reach the United States border without a parent or legal guardian. After apprehension they are first held by U.S. Customs and Border Protection (CBP), whose facilities are designed for short stays. The law generally requires that they be transferred to the Department of Health and Human Services (HHS) within 72 hours, except in exceptional circumstances.

HHS, through its Office of Refugee Resettlement (ORR), cares for the children in shelters and other facilities until they can be released to a vetted sponsor, usually a parent or relative in the United States. The system is therefore a two-stage pipeline:

apprehension, then CBP custody, then transfer to HHS, then HHS care, then discharge to a sponsor.

Arrivals can change sharply from one month to the next, while shelter beds, staff and sponsor-vetting capacity change slowly. When intake runs above discharges for long enough, children accumulate in care, and facilities approach or exceed capacity. The data in this project show how large these swings can be: HHS care rose from 11,551 to 22,557 children in 37 days in spring 2021 (about 297 more children per day), and by October 2026 it had fallen to 1,763.

Analytics can make this pressure visible and measurable. This project turns the daily CBP and HHS figures into capacity and care-load indicators, an early-warning rule, a 60-day forecast and an interactive dashboard that planners can use to see strain before it becomes a crisis.

Data source. HHS Unaccompanied Children Program daily data, published on data.gov / HealthData.gov (catalog.data.gov/dataset/hhs-unaccompanied-alien-children-program). The file used here is the CSV export of that dataset.

2. Problem Statement

The care system for unaccompanied children does not have a simple, data-driven view of how heavily it is loaded, whether a backlog is building, how long children remain in care, and when the next surge is likely to strain capacity. Without that view, shelter capacity and sponsor-processing resources are planned late and reactively.

This project uses the daily CBP and HHS series to (1) measure system load and capacity utilization, (2) compare intake with discharges, (3) estimate length of stay, (4) forecast load for the next 60 days and (5) deliver an interactive Streamlit application with an early-warning rule.

Key questions and the short answers found in this project

Question	Answer from the data	Section
How many children are in the system now?	1,796 on 8 Oct 2026 (33 CBP + 1,763 HHS).	7.2
Is intake outpacing discharges?	Not now: 7-day net intake is -6.0 children/day and the discharge-to-transfer ratio is 1.71. Across 31 Jan 2024 to 8 Oct 2026, intake exceeded discharges on 42% of days.	7.4
How long do children stay?	About 30 days in 2021-2024 (estimate); the same estimate is 123 days now because discharges have become very small.	7.5
When are the peaks?	HHS care peaked at 22,557 on 29 Apr 2021. Within the CBP window the total load peaked at 9,327 on 6 Mar 2024. There is no stable calendar seasonality.	7.3, 7.6
When does the system near capacity?	Utilization is 14% today. At a sustained net intake of +125 children/day (the highest 7-day average seen), the threshold would be reached in about 85 days.	6.10

3. Primary Objectives

#	Objective	How it was met	Section
1	Clean and prepare the daily CBP/HHS data	Removed 10 duplicated dates, 4 isolated spikes and a 71-day gap; built a daily calendar; interpolated stock gaps only.	5
2	Engineer capacity and care-load metrics	Total load, net intake, discharge-to-transfer ratio, 30-day care change, utilization, estimated length of stay, alert flag.	7
3	Analyse trends, seasonality and surges	HHS care trend, month-by-year heatmap, surge periods (2 periods above the threshold).	7.3, 7.6
4	Forecast load and validate it properly	Six models compared on time-based back-tests; ARIMA(7,1,0) chosen; 60-day forecast with an 80% band.	6
5	Build an interactive Streamlit application	Dashboard with KPI cards, five analysis tabs, data-quality tab, sliders and CSV export.	8

4. Secondary Objectives

#	Objective	Result	Section
1	Find the lag between CBP flows and HHS load	Apprehensions turn into transfers about 1 day later (r = 0.58); HHS care reacts the same day to transfers in and discharges out.	7.7
2	Compare surge periods with normal periods	Estimated stay in surge periods: median 31 days vs 35 days outside them. No lengthening during the 2021 surge.	7.5
3	Define and test an early-warning rule	Alert days were followed by care growth in 14 days 70.6% of the time vs 45.6% for all days.	7.8
4	Detect bottlenecks	Estimated CBP custody time rose from 1.3 days (2024) to 3.4 days (2026); HHS turnover slowed sharply from 2025.	7.4, 7.5
5	Provide decision-focused recommendations and a stress test	Surge stress test and five recommendations.	6.10, 10

5. Dataset Description

The dataset is a daily time series with one row per reporting date. The raw file has **1,366 rows and 6 columns**, covering 23 Mar 2021 to 8 Oct 2026 after cleaning. It mixes **stock** variables (children present at the end of the day) and **flow** variables (children moving during the day).

Original column	Short name	Type	Meaning	Available
Date	Date	Date	Reporting date	All
Children apprehended and placed in CBP custody*	cbp_in	Flow	New arrivals entering CBP custody	From 31 Jan 2024
Children in CBP custody	cbp_custody	Stock	Children held by CBP at day end	From 31 Jan 2024
Children transferred out of CBP custody	cbp_transfer	Flow	Moved from CBP to HHS	From 31 Jan 2024
Children in HHS Care	hhs_care	Stock	Children in HHS care at day end	All
Children discharged from HHS Care	hhs_discharge	Flow	Released to sponsors that day	All

Data quality findings and how each was handled

Finding in the raw file	Size	Treatment
CBP columns are empty before 31 Jan 2024	701 of 1,366 rows	CBP-based metrics (total load, flows, ratio, alerts, lags) are computed only from 31 Jan 2024. HHS metrics cover the whole period.
Duplicated dates with conflicting values	10 dates, 2022-01-02 to 2022-01-26; HHS care differs by up to 1,480	Two interleaved series appear in January 2022. The first-listed row was kept because it continues smoothly from the days before and after.
Isolated spikes in HHS care	4 values: 15 Jul 2021 (19,920), 8 Jan 2022 (7,340), 11 Jan 2022 (6,850), 16 Jan 2022 (6,351)	Values that differ by more than 12% from the surrounding 9-day median, with neighbours within 4 days, were treated as errors and re-filled by interpolation.
Very long gap at the start	71 days (11 Jan 2021 to 23 Mar 2021)	The single earlier row was dropped; analysis starts on 23 Mar 2021. Interpolating across 71 days would invent data.
Reporting only Sunday to Thursday	Sun 247, Mon 277, Tue 281, Wed 275, Thu 271, Fri 3, Sat 1	671 calendar days have no report (longest run 13 days). Stocks are interpolated across these days; flows are never interpolated and are averaged over reported days only.
Flows do not add up to the change in HHS care	Cumulative net flow 902 vs change in care -6,419 since 31 Jan 2024	Day to day they move together (correlation 0.89 on 516 consecutive days), but sums drift. Backlog is therefore measured from the stock (HHS care), not by adding flows.

The asterisk on the apprehension column refers to a footnote in the source data, so that column should be read as reported by the source rather than as an exact count of unique children.

Summary by year (HHS care on the full daily calendar; discharges per reported day)

Year	Average HHS care	Minimum	Maximum	Average discharges / day
2021	14,642	10,540	22,557	432.6
2022	9,853	7,924	11,977	320.9
2023	8,595	5,679	11,516	288.0
2024	7,020	5,664	10,870	205.9
2025	2,535	1,972	6,405	28.6
2026	2,003	1,757	2,466	11.9

2021 starts on 23 March and 2026 ends on 8 October.

6. Forecasting Methodology (Step-by-Step)

Step 1 - Define the targets and horizon. The primary target is the number of children in HHS care (full period). The secondary target is total system load, CBP custody plus HHS care, which exists from 31 Jan 2024. The horizon is 60 days.

Step 2 - Prepare the series. A full daily calendar was built; gaps in stock series were filled by linear interpolation (671 unreported days, longest run 13 days); duplicates, spikes and the 71-day gap were handled as in Section 5.

Step 3 - Examine the patterns. HHS care is not stationary: it moves between 22,557 and 1,757 with long swings and level shifts, and has no stable weekly pattern (weekend values are interpolated). Models were therefore built on first differences (day-to-day changes), with no seasonal terms.

Step 4 - Validate in time order, never at random. Two tests were used: (a) the final 60 days held out; (b) a rolling-origin test with 12 forecast origins spaced 30 days apart, each forecasting 30 days ahead (covering the last year). Rolling origins are more reliable than one split because performance changes with the regime.

Step 5 - Baselines. Naive (repeat the last value), 7-day moving average, and drift (extend the last 30-day slope). Any model must beat these.

Step 6 - Fit models. ARIMA(p,1,0) for p = 1, 3 and 7, that is, an autoregression on the daily change fitted by least squares on the most recent 365 days, with no drift term (in preliminary tests a drift term made the 30-day errors much larger for p = 3 and no better for p = 7). Implemented in NumPy in *uac_core.py*, so the report and the Streamlit app give the same numbers.

Step 7 - Evaluate. MAE (children) and MAPE (%) for each model, shown below.

HHS care

Model	Rolling-origin MAE	MAPE	Last-60-days MAE	MAPE
Naive (last value)	52.6	2.58%	29.6	1.63%
7-day moving average	63.6	3.10%	35.8	1.98%
Drift (last 30 days)	33.7	1.64%	48.7	2.69%
ARIMA(1,1,0)	47.5	2.34%	28.6	1.57%
ARIMA(3,1,0)	41.1	2.00%	27.7	1.52%
ARIMA(7,1,0)	**31.1**	1.54%	31.5	1.70%

Total system load (from 31 Jan 2024)

Model	Rolling-origin MAE	MAPE	Last-60-days MAE	MAPE
Naive (last value)	50.4	2.42%	28.9	1.56%
7-day moving average	60.3	2.90%	35.4	1.92%
Drift (last 30 days)	**30.4**	1.46%	42.9	2.33%
ARIMA(1,1,0)	46.6	2.26%	28.8	1.56%
ARIMA(3,1,0)	42.0	2.02%	28.5	1.54%
ARIMA(7,1,0)	37.0	1.76%	32.6	1.74%

Lowest rolling-origin MAE in bold. Lower is better.

Step 8 - Select the model. Among the ARIMA models, ARIMA(7,1,0) has the lowest rolling-origin error. For HHS care it is also the best of all six models there (MAE 31.1 against 52.6 for the naive baseline, a reduction of 41%). Two honest caveats: on the single final 60-day hold-out all models are close (MAE between 28 and 49) and the naive baseline is slightly better than ARIMA(7,1,0); and for total load the 30-day drift baseline has the lowest rolling error (30.4), so ARIMA(7,1,0) is used there for consistency.

Step 9 - Refit and forecast. The chosen model is refitted on all data and run 60 days ahead. The 80% band comes from 1,000 simulated paths that resample the model's own past errors.

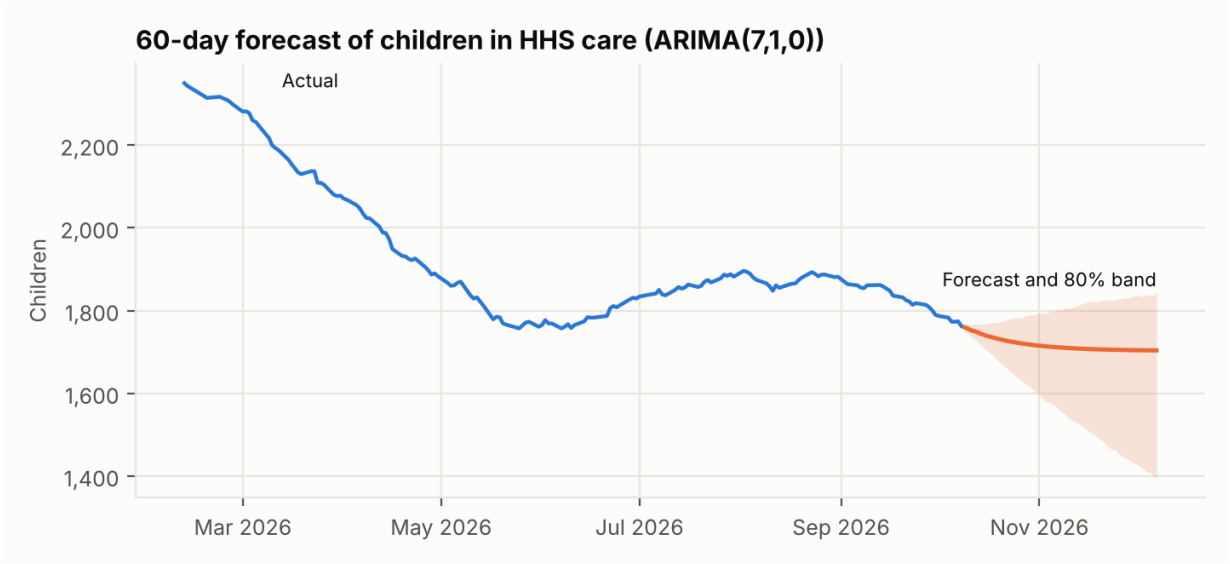


Figure 6.1 - Forecast of children in HHS care to 7 Dec 2026. The band widens with the horizon.

Target	Last actual	Day 30	Day 60 (7 Dec 2026)
HHS care	1,763	1,711 (1,560 to 1,802)	1,704 (1,395 to 1,844)
Total system load	1,796	1,779 (1,635 to 1,857)	1,779 (1,535 to 1,895)

Step 10 - Interpret in capacity terms. The 60-day forecast for HHS care is 1,704 children, 14% of the 12,336 threshold; even the top of the band (1,844) is 15%. On current trends no capacity problem is expected in the next 60 days. A statistical forecast cannot anticipate a sudden change in arrivals, so it is paired with a stress test:

Sustained net intake (children/day)	HHS care after 60 days	Utilization after 60 days	Days to reach the threshold
+25	3,263	26%	423
+50	4,763	39%	211
+100	7,763	63%	106
+125	9,263	75%	85

Starting from 1,763 children and a threshold of 12,336. +125/day is the highest 7-day average net intake observed (29 Apr 2024). In spring 2021 HHS care rose by about 297 children per day for 37 days, which is much faster than any of these scenarios.

7. Key Performance Indicators (KPIs)

7.1 KPI definitions

KPI	Formula	How to read it
Total system load	CBP custody + HHS care	All children in the system (from 31 Jan 2024)
Net intake (daily)	Transfers out of CBP - HHS discharges	Positive: HHS load tends to rise
7-day net intake	Average daily net intake over the last 7 days (reported days only)	Smooths day-to-day noise
Discharge-to-transfer ratio (7-day)	Sum of discharges / sum of transfers over 7 days	Below 1: intake exceeds discharges
Backlog indicator	30-day change in HHS care	Positive: backlog growing (measured from the stock)
Capacity utilization	HHS care / 12,336 (assumed capacity)	Above 100%: over the threshold
Estimated length of stay	HHS care / 30-day average daily discharges (Little's law)	Approximate days in care
Estimated CBP custody time	Average CBP custody / average daily transfers	Approximate days in CBP custody
Alert	7-day net intake > 0 and 7-day ratio < 0.9	Early-warning signal

Capacity assumption. The data contain no official capacity. The threshold used is the 90th percentile of HHS care over the whole period (12,336 children). For comparison, the 80th percentile is 10,630 (404 days above) and the 95th is 14,941 (102 days above). The Streamlit app lets the user change this value.

7.2 Snapshot on 8 Oct 2026

KPI	Value
Total system load	1,796 children
Children in CBP custody	33
Children in HHS care	1,763
7-day net intake	-6.0 children per day
7-day discharge-to-transfer ratio	1.71
Capacity utilization	14.3%
Estimated length of stay in HHS care	123 days
Early-warning alert	Off (last alert day: 22 Aug 2026)

7.3 Load and capacity

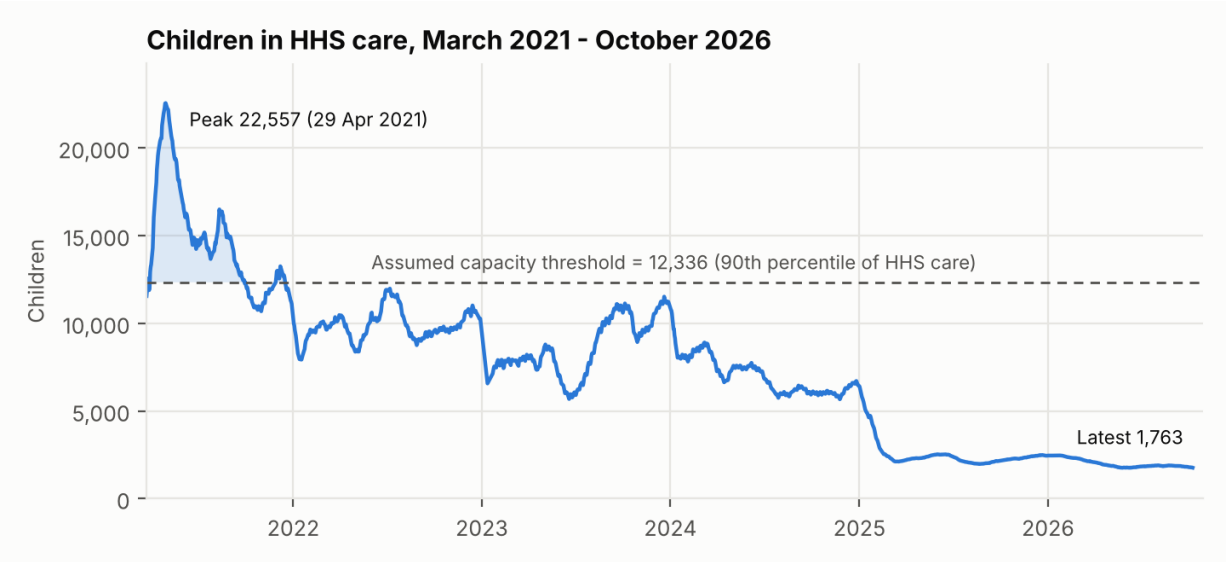


Figure 7.1 - Children in HHS care and the assumed capacity threshold. Shading shows the days above the threshold.

HHS care peaked at 22,557 on 29 Apr 2021, 183% of the threshold. It was above the threshold on 203 days (10.0% of the period), in two spells: 30 Mar 2021 to 28 Sep 2021 (183 days) and 29 Nov 2021 to 16 Dec 2021 (18 days). After 2021 the load stayed below the threshold. It fell step by step to about 7,020 on average in 2024, then dropped sharply in early 2025 (from 6,430 on 31 Dec 2024 to 2,109 by 18 Mar 2025), and has stayed near 2,003 since. The total system load peaked at 9,327 on 6 Mar 2024 and is 1,796 today.

7.4 Intake versus discharge

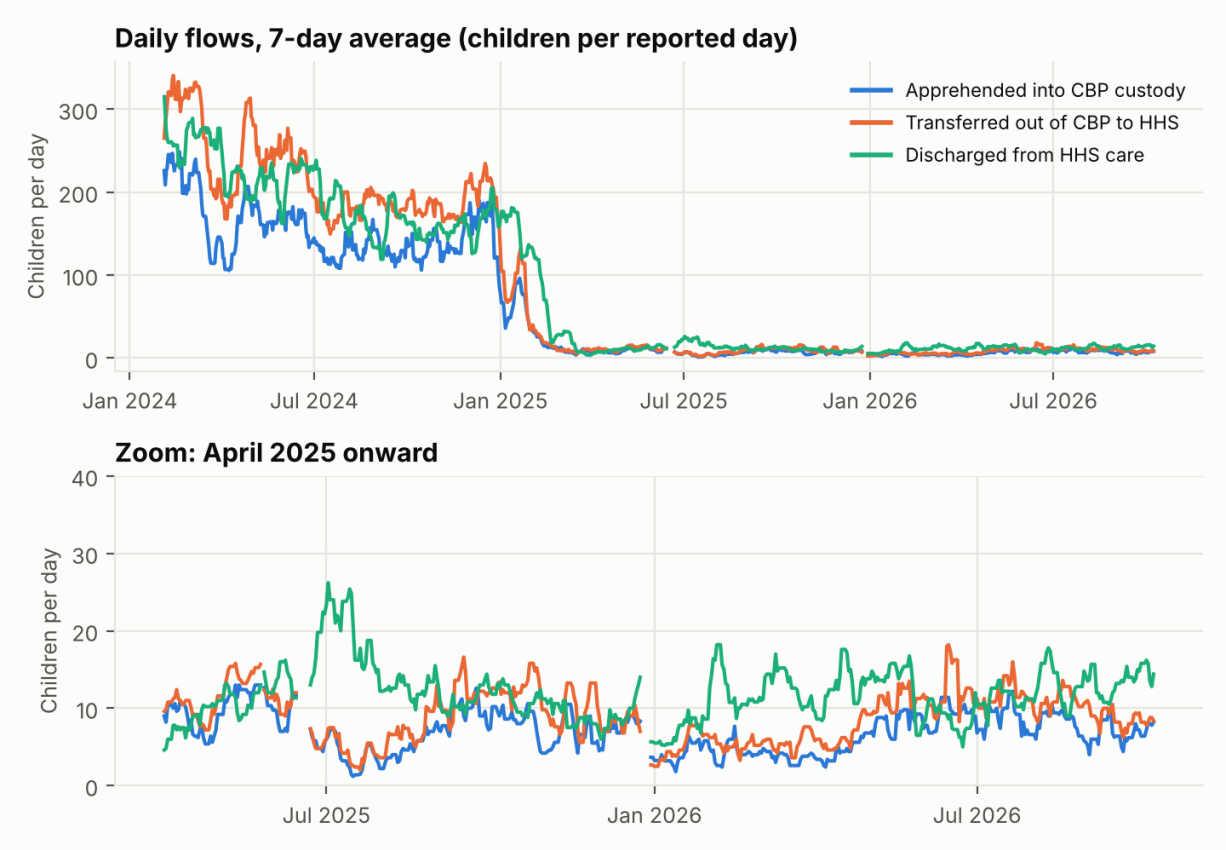


Figure 7.2 - Daily flows (7-day averages). The lower panel zooms in on April 2025 onward, where flows are only 5 to 25 children a day.

Year	Apprehended / day	Transferred to HHS / day	Discharged / day	Average CBP custody	Est. CBP custody time (days)
2024	153.4	217.2	197.6	285.7	1.3
2025	12.9	16.4	28.6	32.3	2.0
2026	6.5	8.6	11.9	29.6	3.4

Averages over reported days. 2024 starts on 31 January and 2026 ends on 8 October.

Apprehensions fell from 153 a day in 2024 to 12.9 in 2025 and 6.5 in 2026. The busiest day in the CBP window was 15 Feb 2024 with 333 apprehensions, and CBP custody peaked at 531 on 4 Feb 2024. The yearly median of the 7-day discharge-to-transfer ratio was 0.89 in 2024 (below 1, intake ahead of discharges), 1.22 in 2025 and 1.44 in 2026. Intake exceeded discharges on 42% of days; the longest unbroken spell lasted 58 days (24 Sep 2024 to 20 Nov 2024), and the highest 7-day net intake was +125 children a day (29 Apr 2024).

The estimated time in CBP custody (average custody divided by average transfers) rose from 1.3 days in 2024 to 3.4 days in 2026. This is an approximation, but it points to slower transfers out of CBP now that volumes are small.

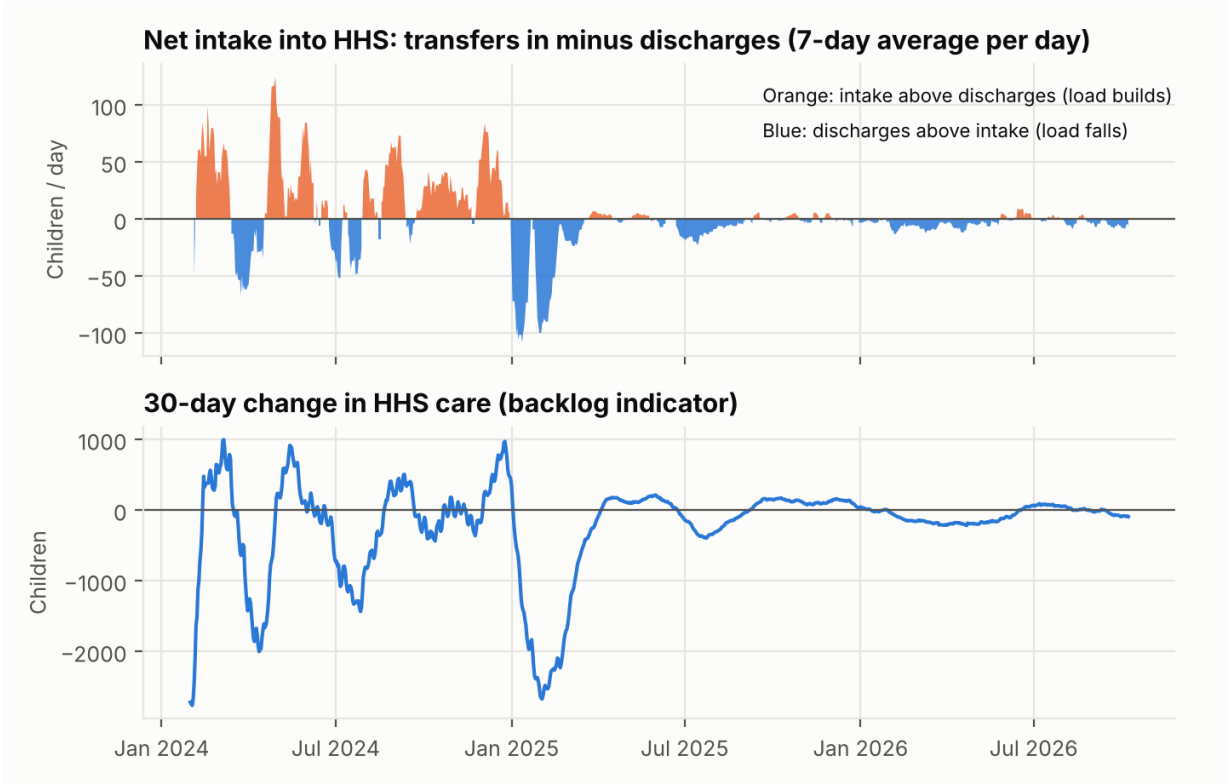


Figure 7.3 - Net intake into HHS (top) and the 30-day change in HHS care, the backlog indicator (bottom).

Caution. Net intake calculated from the flows overstates the actual change in care in 2024: cumulative net intake was 4,501 children while HHS care changed by -1,728. Flows are reported on about five days in seven, and the data do not say what other movements are missing. The 7-day net intake and the ratio are therefore used as relative signals, and the backlog is read from the stock.

7.5 Estimated length of stay

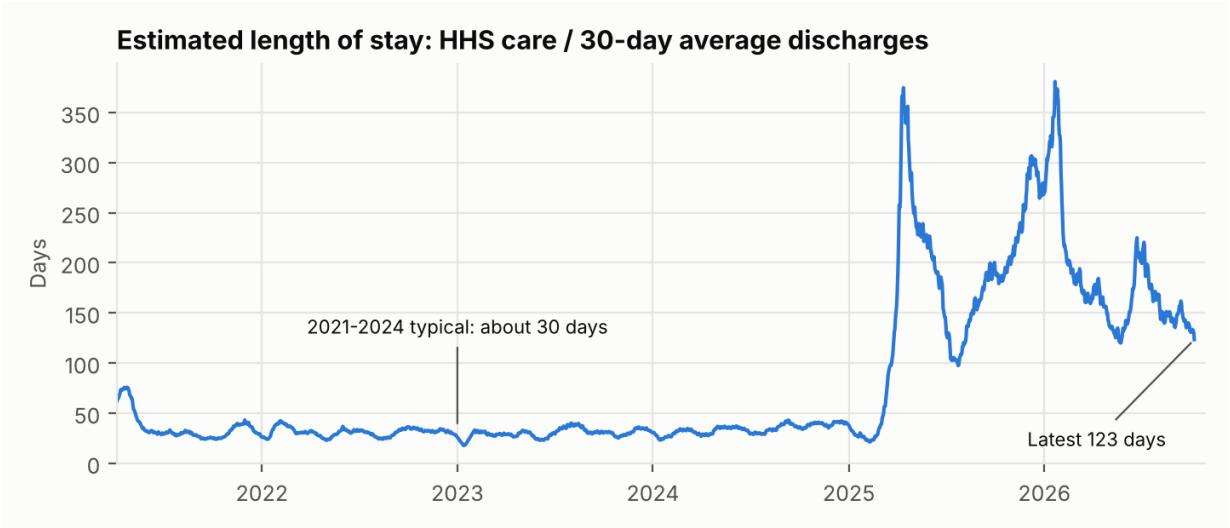


Figure 7.4 - Estimated length of stay (HHS care divided by the 30-day average of daily discharges).

From 2021 to 2024 the estimate is steady at about 30 days (yearly medians 31, 31, 30 and 34 days). It then jumps to 182 days (2025 median) and 164 days (2026), with a high of 381 days on 21 Jan 2026; the latest value is 123 days. The jump comes from discharges falling from about 288 a day in 2023 to 11.9 in 2026 while 2,003 children remain in care.

Surge versus normal. During the days above the capacity threshold the median estimate was 31 days, against 35 days outside those periods. So the 2021 surge did not lengthen stays; the system moved more children through, in step with arrivals. The change that matters is the slow turnover since 2025.

Interpretation. Dividing care by discharges is a rough estimate that assumes the system is in a steady state. Since 2025 it is not, and the children who remain are likely to include many long stays. The figure should be read as a signal that turnover has slowed sharply, not as the exact average stay.

7.6 Seasonality

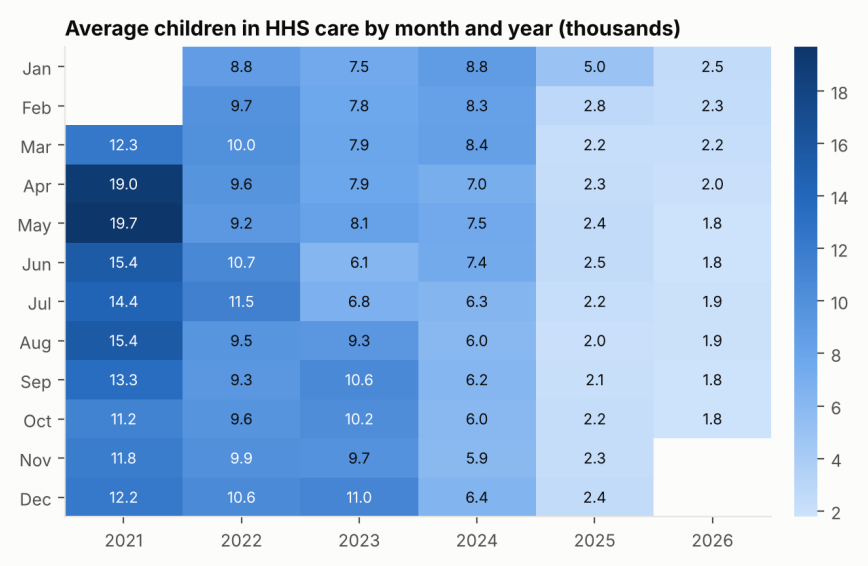


Figure 7.5 - Average children in HHS care by month and year (thousands).

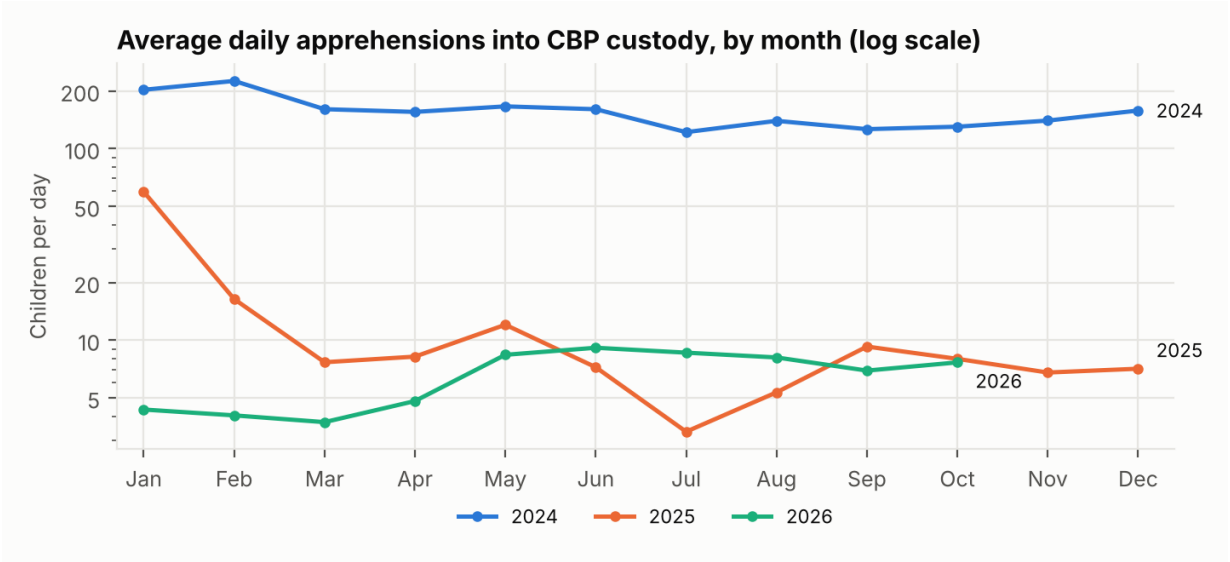


Figure 7.6 - Average daily apprehensions into CBP custody by calendar month, 2024 to 2026 (log scale).

The heatmap reads down the years rather than across the months: the colour changes far more from one year to the next than from one month to the next. Apprehension rates also differ by an order of magnitude between 2024, 2025 and 2026, so with three years of CBP data and such a large level shift no stable calendar seasonality can be established. The months with the highest averages in the CBP window (December, February, November) mostly reflect the high 2024 level and should not be treated as a seasonal peak.

7.7 Lag between CBP flows and HHS load

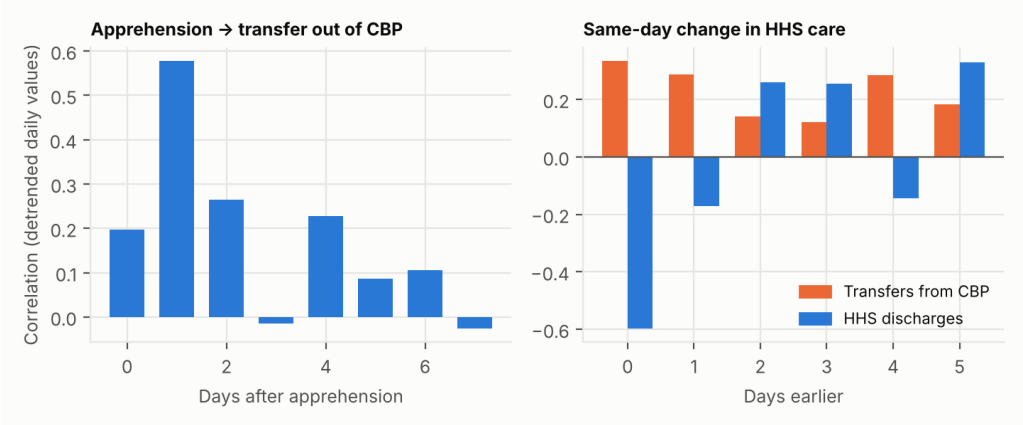


Figure 7.7 - Correlations after removing each series' 31-day moving average, so that large level changes between years do not create false correlation.

Apprehensions are followed by transfers out of CBP mostly one day later (correlation 0.58, with a smaller effect after two days). HHS care responds on the same day: the day's change in care is positively related to transfers (r = 0.33) and negatively to discharges (r = -0.60). In practice the planning lead time between a rise in apprehensions and a rise in HHS load is only a day or two, so early warning has to come from the trend in flows over several days and from the stock, not from waiting for the lag.

7.8 Early-warning rule

Rule: raise an alert when the 7-day average net intake is above zero **and** the 7-day discharge-to-transfer ratio is below 0.9. The rule was tested on 951 days with enough history: did HHS care grow over the following 14 days?

Ratio threshold	Alert days	Followed by growth in 14 days	Average 14-day change after alert	Average change on other days
0.8	181	82.9%	+77	-133
0.9 (used)	296	70.6%	+36	-152
1.0	411	65.0%	-5	-160
All days (base rate)	951	45.6%		

With the 0.9 threshold the rule gave 296 alert days (172 in 2024, 82 in 2025, 42 in 2026; the last on 22 Aug 2026). Alert days were followed by growth in HHS care 70.6% of the time, against 45.6% for all days, and the average 14-day change was +36 children after an alert against -152 otherwise. A stricter threshold (0.8) is more reliable but fires less often. The thresholds were not tuned to the data, and consecutive days are not independent, so these figures show that the rule carries information, not its exact accuracy.

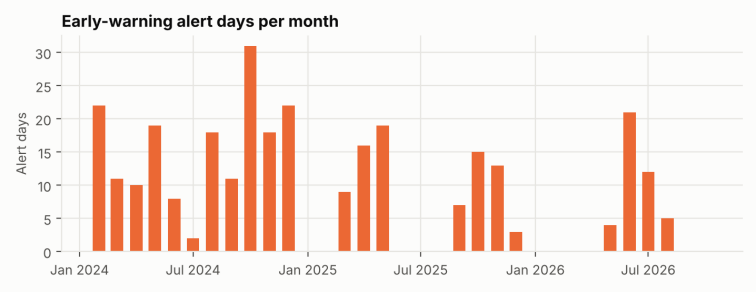


Figure 7.8 - Alert days per month.

8. Streamlit Web Application Requirements

The web application is the interactive front end of the analysis. It reads the raw CSV, applies the same cleaning, metric and forecasting functions as this report (module *uac_core.py*), and presents the results to a non-technical user.

Functional requirements

Requirement	Implementation
Data input	Loads data/uac.csv by default; optional CSV upload in the sidebar. Cleaning is automatic and every step is reported in the Data quality tab.
Interactive controls	Date-range selector; capacity-percentile slider; forecast horizon (14 to 90 days); forecast target (HHS care or total load); ARIMA order; stress-test slider.
KPI cards	Total system load, HHS care, 7-day net intake, estimated length of stay, capacity utilization, early-warning status.
Trends tab	HHS care with the capacity line; CBP custody and total load; estimated length of stay.
Intake vs Discharge tab	7-day flows, 7-day net intake, 30-day change in HHS care.
Seasonality tab	Month-by-year heatmap; apprehensions by month and year.
Forecast tab	Forecast with 80% band, day-N value and utilization, surge stress test.
Alerts tab	Alert days in the range, monthly bar chart and table.
Export	Download of the processed data as CSV.

Non-functional requirements

- Python 3.9 or newer; dependencies in requirements.txt (streamlit, pandas, numpy, plotly).
- Results are cached (st.cache_data) so charts respond quickly after the first load.
- Clear titles, labelled axes, and a caption that states the capacity assumption and the data limits.
- Deployable on Streamlit Community Cloud from a GitHub repository (no secrets needed).

How to run

```
pip install -r requirements.txt
streamlit run app.py
```

Files

File	Role
app.py	Streamlit interface (listing in Appendix A)
uac_core.py	Cleaning, metrics and forecasting functions shared by the app and the analysis
analysis.py	Full analysis script: produces results.json and all charts (can be pasted into Jupyter cells)
data/	uac.csv (raw data) and uac_clean_with_metrics.csv (cleaned, with all KPI columns)

Verification. The data, KPI and forecast code behind the app was run on the provided dataset; its KPI values (total load 1,796, HHS care 1,763, utilization 14%) match Section 7.2.

9. Deliverables and Submission

#	Deliverable	File / location	Status
1	Project report (this document)	UAC_Capacity_Care_Load_Project_Report.pdf	Done
2	Streamlit application	app.py, uac_core.py, requirements.txt	Done
3	Analysis code (cleaning, KPIs, EDA, forecast)	analysis.py	Done
4	Cleaned dataset with KPI columns	data/uac_clean_with_metrics.csv	Done
5	Raw dataset	data/uac.csv	Done
6	Charts used in the report	charts/ (9 PNG files)	Done
7	GitHub repository link	_____	To add
8	Deployed app link (Streamlit Community Cloud)	_____	To add

Repository structure

```
uac-capacity-analytics/
|-- app.py
|-- uac_core.py
|-- analysis.py
|-- requirements.txt
|-- README.md
|-- data/
|   |-- uac.csv
|   `-- uac_clean_with_metrics.csv
|-- charts/            (9 PNG files)
`-- report/
    `-- UAC_Capacity_Care_Load_Project_Report.pdf
```

Steps to publish

- Create a GitHub repository and upload the folder above.
- On share.streamlit.io choose New app, select the repository, branch main and file app.py, then Deploy.
- Add the repository link and the app link in the table above, and submit them with this report.

10. Conclusion

The project turned the daily CBP and HHS series into a working picture of system capacity and care load, and into a dashboard that can be used for monitoring. The main findings are:

- **Load.** HHS care peaked at 22,557 on 29 Apr 2021 (183% of the assumed threshold) and was above the threshold on 203 days, all in 2021. It is now 1,763 (14% of the threshold); with CBP custody the system holds 1,796 children.
- **Intake and discharge.** Apprehensions fell from 153 a day in 2024 to 6.5 in 2026. Intake exceeded discharges on 42% of days since 2024, but at present discharges are ahead (7-day ratio 1.71).
- **Length of stay.** About 30 days in 2021-2024 and no increase during the 2021 surge. Since 2025 the estimate is far higher (123 days now, peaking at 381) because discharges are only about 12 a day: turnover has slowed sharply.
- **Timing.** Transfers follow apprehensions by about one day and HHS care reacts the same day, so the warning time is short. No stable calendar seasonality exists in the data.
- **Forecast.** ARIMA(7,1,0) gives 1,704 children in HHS care on 7 Dec 2026 (80% band 1,395 to 1,844), about 14% of the threshold. It cut the 30-day error by 41% against the naive baseline in rolling tests.
- **Early warning.** The alert rule (7-day net intake above zero and ratio below 0.9) was followed by growth in HHS care within 14 days 70.6% of the time, against 45.6% for all days.

Recommendations

- **Keep the early-warning rule in daily use** and review it weekly; it carries real signal (Section 7.8), and a ratio threshold of 0.8 can be used when fewer, stronger alerts are wanted.
- **Plan surge readiness around speed, not season.** HHS care once grew by about 297 children a day for 37 days, and at +125 a day the threshold would be reached in about 85 days. Keep a plan to add capacity within weeks, and trigger it from the alert and the 30-day change in care.
- **Examine the slow turnover.** With about 12 discharges a day and 1,763 children in care, review the long-stay cohort and the sponsor-vetting and placement process.
- **Watch the CBP stage.** The estimated custody time rose from 1.3 to 3.4 days; track transfers against the 72-hour expectation.
- **Improve the data.** Publish flows for all seven days, remove duplicated dates, extend the CBP columns back before 2024, and state an official capacity figure so utilization can be measured exactly.

Limitations

- The data are daily aggregates: no individual cases, ages, countries of origin or facility locations can be analysed.
- Capacity is an assumption (90th percentile of HHS care), not an official number; conclusions about 'over capacity' depend on it.
- CBP-based results cover only 31 Jan 2024 onward, and flows are reported on about five days in seven; flows do not fully reconcile with changes in care.
- Length of stay from Little's law is an approximation and is least reliable when the system is shrinking, as since 2025.

- The forecast cannot foresee policy changes or sudden surges; the stress test is a scenario, not a prediction. The alert evaluation uses 2024 to 2026 only and consecutive days are not independent.

Future work: add official capacity and facility data, break the analysis down by age and region if available, include external drivers, and test machine-learning models on lagged features.

Sources

HHS Unaccompanied Children Program dataset, data.gov: catalog.data.gov/dataset/hhs-unaccompanied-alien-children-program (CSV export used for this project).

Appendix A - Streamlit application listing (app.py)

```
"""Streamlit dashboard - System Capacity & Care Load Analytics (Unaccompanied Children).
Run:  streamlit run app.py
"""
import os
import numpy as np
import pandas as pd
import plotly.express as px
import plotly.graph_objects as go
import streamlit as st

from uac_core import load_clean, add_metrics, forecast_ar_diff

HERE = os.path.dirname(os.path.abspath(__file__))
DEFAULT_CSV = os.path.join(HERE, "data", "uac.csv")

st.set_page_config(page_title="UAC Capacity & Care Load", layout="wide")
st.title("System Capacity & Care Load Analytics - Unaccompanied Children")
st.caption("CBP custody and HHS care: load, intake vs discharge, length of stay, early warning and forecast.")


@st.cache_data(show_spinner=False)
def get_data(source, cap_pct):
    df, log = load_clean(source)
    d, cap = add_metrics(df, cap_pct)
    return d, cap, log


@st.cache_data(show_spinner=False)
def get_forecast(series, horizon, p):
    return forecast_ar_diff(series, horizon, p=p, window=365, n_sim=400, seed=1)


# ---------------------------------------------------------------- sidebar
st.sidebar.header("Controls")
up = st.sidebar.file_uploader("Upload dataset (CSV) - optional", type="csv")
if up is None and not os.path.exists(DEFAULT_CSV):
    st.info("Upload the dataset CSV from the sidebar to begin.")
    st.stop()
cap_pct = st.sidebar.slider("Capacity threshold (percentile of HHS care)", 50, 99, 90)
d, cap, log = get_data(up if up is not None else DEFAULT_CSV, cap_pct)

rng = st.sidebar.date_input("Date range", [d.index.min().date(), d.index.max().date()],
                            min_value=d.index.min().date(), max_value=d.index.max().date())
if len(rng) != 2:
    st.stop()
view = d.loc[str(rng[0]):str(rng[1])]
horizon = st.sidebar.slider("Forecast horizon (days)", 14, 90, 60)
target = st.sidebar.selectbox("Forecast target", ["HHS care", "Total system load (CBP + HHS, from 2024)"])
order_p = st.sidebar.selectbox("ARIMA order p (ARIMA(p,1,0))", [1, 3, 7], index=2)

# ---------------------------------------------------------------- KPI cards
def last_valid(col):
    s = view[col].dropna()
    return s.iloc[-1] if len(s) else np.nan

def show(v, fmt):
    return "n/a" if pd.isna(v) else format(v, fmt)

k = st.columns(6)
k[0].metric("Total system load", show(last_valid("total_load"), ",.0f"))
k[1].metric("Children in HHS care", show(last_valid("hhs_care"), ",.0f"))
k[2].metric("7-day net intake / day", show(last_valid("net_7d"), "+.1f"))
k[3].metric("Est. length of stay (days)", show(last_valid("est_stay_days"), ".0f"))
k[4].metric("Capacity utilization", show(last_valid("utilization"), ".0%"))
alert_now = bool(view["alert"].iloc[-1])
k[5].metric("Early-warning alert", "ON" if alert_now else "off")
```

```
st.caption(f"Assumed capacity threshold = {cap:,.0f} children (percentile {cap_pct} of HHS care, whole dataset). "
           "CBP-based metrics exist only from 31 Jan 2024.")

t1, t2, t3, t4, t5, t6 = st.tabs(["Trends", "Intake vs Discharge", "Seasonality", "Forecast", "Alerts", "Data quality"])

with t1:
    fig = px.line(view, y="hhs_care", labels={"hhs_care": "Children in HHS care", "Date": ""}, title="Children in HHS care")
    fig.add_hline(y=cap, line_dash="dash", annotation_text="capacity threshold")
    st.plotly_chart(fig, use_container_width=True)
    c = view.dropna(subset=["cbp_custody"])
    if len(c):
        st.plotly_chart(px.line(c, y=["cbp_custody", "total_load"], title="CBP custody and total system load (from 2024)"),
                        use_container_width=True)
    st.plotly_chart(px.line(view, y="est_stay_days", title="Estimated length of stay (days) = HHS care / 30-day avg discharges"),
                    use_container_width=True)

with t2:
    c = view.dropna(subset=["cbp_custody"])
    if len(c):
        flows = c[["cbp_in", "cbp_transfer", "hhs_discharge"]].rolling("7D", min_periods=3).mean()
        st.plotly_chart(px.line(flows, title="Flows, 7-day average (children per reported day)"), use_container_width=True)
        st.plotly_chart(px.area(c, y="net_7d", title="Net intake into HHS (transfers in - discharges), 7-day average"),
                        use_container_width=True)
        st.plotly_chart(px.line(c, y="care_change_30d", title="30-day change in HHS care (backlog indicator)"),
                        use_container_width=True)
    else:
        st.info("CBP flow data is not available in the selected range (it starts on 31 Jan 2024).")

with t3:
    p = view.assign(year=view.index.year, month=view.index.month).pivot_table(
        values="hhs_care", index="month", columns="year", aggfunc="mean")
    st.plotly_chart(px.imshow(p, aspect="auto", color_continuous_scale="Blues", text_auto=".0f",
                              title="Average children in HHS care: month x year"), use_container_width=True)
    c = view.dropna(subset=["cbp_in"])
    if len(c):
        m = c.assign(year=c.index.year, month=c.index.month).groupby(["year", "month"])["cbp_in"].mean().reset_index()
        m["year"] = m["year"].astype(str)
        st.plotly_chart(px.line(m, x="month", y="cbp_in", color="year", markers=True, log_y=True,
                                title="Average daily apprehensions by month (log scale)"), use_container_width=True)

with t4:
    col = "hhs_care" if target == "HHS care" else "total_load"
    series = d[col].dropna()
    fc = get_forecast(series, horizon, order_p)
    hist = series.asfreq("D").interpolate().iloc[-240:]
    fig = go.Figure()
    fig.add_scatter(x=hist.index, y=hist, name="Actual")
    fig.add_scatter(x=fc.index, y=fc["mean"], name="Forecast")
    fig.add_scatter(x=fc.index, y=fc["lower"], line=dict(width=0), showlegend=False, hoverinfo="skip")
    fig.add_scatter(x=fc.index, y=fc["upper"], fill="tonexty", line=dict(width=0), name="80% band")
    fig.update_layout(title=f"{horizon}-day forecast - {target} (ARIMA({order_p},1,0))")
    st.plotly_chart(fig, use_container_width=True)
    st.write(f"Forecast on day {horizon}: **{fc['mean'].iloc[-1]:,.0f}** children "
             f"(80% band {fc['lower'].iloc[-1]:,.0f} - {fc['upper'].iloc[-1]:,.0f}); "
             f"utilization of the assumed threshold: **{fc['mean'].iloc[-1] / cap:.0%}**.")
    st.markdown("**Surge stress test** - days until the capacity threshold at a sustained net intake")
    cur = d["hhs_care"].iloc[-1]
    rate = st.slider("Sustained net intake (children per day)", 10, 200, 125)
    st.write(f"At +{rate} children/day, HHS care would reach the threshold in about "
             f"**{(cap - cur) / rate:,.0f} days** (from {cur:,.0f} today).")

with t5:
    a = view[view["alert"]]
    st.write(f"Alert days in selected range: **{len(a)}** of {len(view)}")
    st.caption("Rule: 7-day average net intake > 0 AND 7-day discharge/transfer ratio < 0.9")
    if len(a):
        monthly = a.resample("MS").size().rename("alert_days").reset_index()
        st.plotly_chart(px.bar(monthly, x="Date", y="alert_days", title="Alert days per month"), use_container_width=True)
        st.dataframe(a[["hhs_care", "net_7d", "dis_ratio_7d", "utilization"]].round(2))

with t6:
    st.write("How the raw file was cleaned:")
    st.json({k_: v for k_, v in log.items()})

st.download_button("Download processed data (CSV)", view.to_csv().encode(), "uac_processed.csv")
```